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324732(24)

B. E. (Seventh Semester) Examination, April-May, 2016

(New Scheme)

(Elect. Engg. Branch)

MODERN CONTROL SYSTEM

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all the questions. Part (a) from each question is compulsory. Attempt any two parts from parts (b), (c) and (d) of each question.*

Unit-I

1. (a) What are stable and unstable limit cycles? 2

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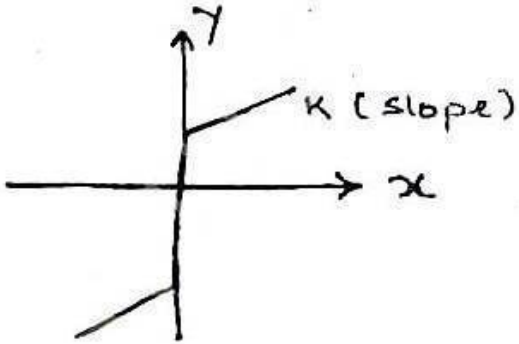
(b) Differentiate between linear and non linear control system. 7

(c) What are describing functions? Derive the describing function of a relay with dead zone. 7

(d) Determine the describing function of the following non-linearity. 7



Unit-II

2. (a) Define state and state variable. 2

(b) What is state transition matrix? State the properties of STM. 7

(c) Option state space representation of the spring mass damper system. 7

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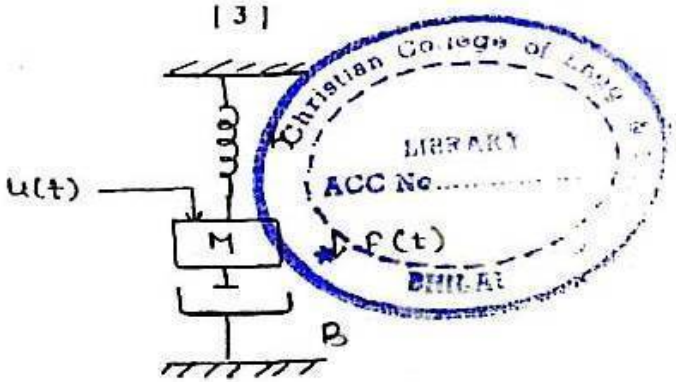

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(d) A system characterised by the transfer function

$$\frac{y(s)}{u(s)} = \frac{2}{s^3+6s^2+11s+6}$$

Find the state and output equation in matrix form and also test the controllability and observability. 7

Unit-III

3. (a) Define positive and negative definiteness. 2

(b) State and prove Liapunov's theorem of stability for linear system. 7

(c) Determine the stability using Lapunov's method for system represented by

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x.$$ 7

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(d) Explain the method of constructing Liapunov function using Krasovskii's method. 7

Unit-IV

4. (a) Write the necessary and sufficient condition for arbitrary pole placement. 2

(b) Derive Ackermann's formula for the determination of state feedback gain matrix 'K'. 7

(c) Explain the effect of the addition of the observes on a closed loop system. 7

(d) Consider the system

$$\dot{x} = Ax + Bu$$
$$y = cx$$

where

$$A = \begin{bmatrix} 0 & 20{\cdot}6 \\ 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

Design a full order state observer by using direct substitution approach. 7

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Unit-V

5. (a) What is zero order hold? 2

(b) Explain the stability analysis by the use of the bilinear transformation. 7

(c) State and explain the optimal control problem and performance index for optimal control problem. 7

(d) Explain Impulse Sampling and data hold circuit. 7

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